

IBM ECM System Monitor Field Guides

Integration with Grafana using Infinity data source

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Introduction

Overview

This guide describes in detail how IBM ECM System Monitor (ESM) can be integrated into Grafana to review ESM incidents.

Disclaimer

The content of this document is based on ESM Version 5.7.0.0-000. The descriptions and guidelines in this document are for informational purposes only. Up-to-dateness, content completeness, appropriateness and validity for all possible scenarios cannot be guaranteed. All information is provided on an as-is basis. The author is not liable for any errors or omissions in this document or any losses, injuries and damage arising from its use.

If you are planning to set up or configure ESM or to adjust an existing installation, it is necessary to consider current security whitepapers, release notes and announcements from the official IBM ECM System Monitor product documentation website.

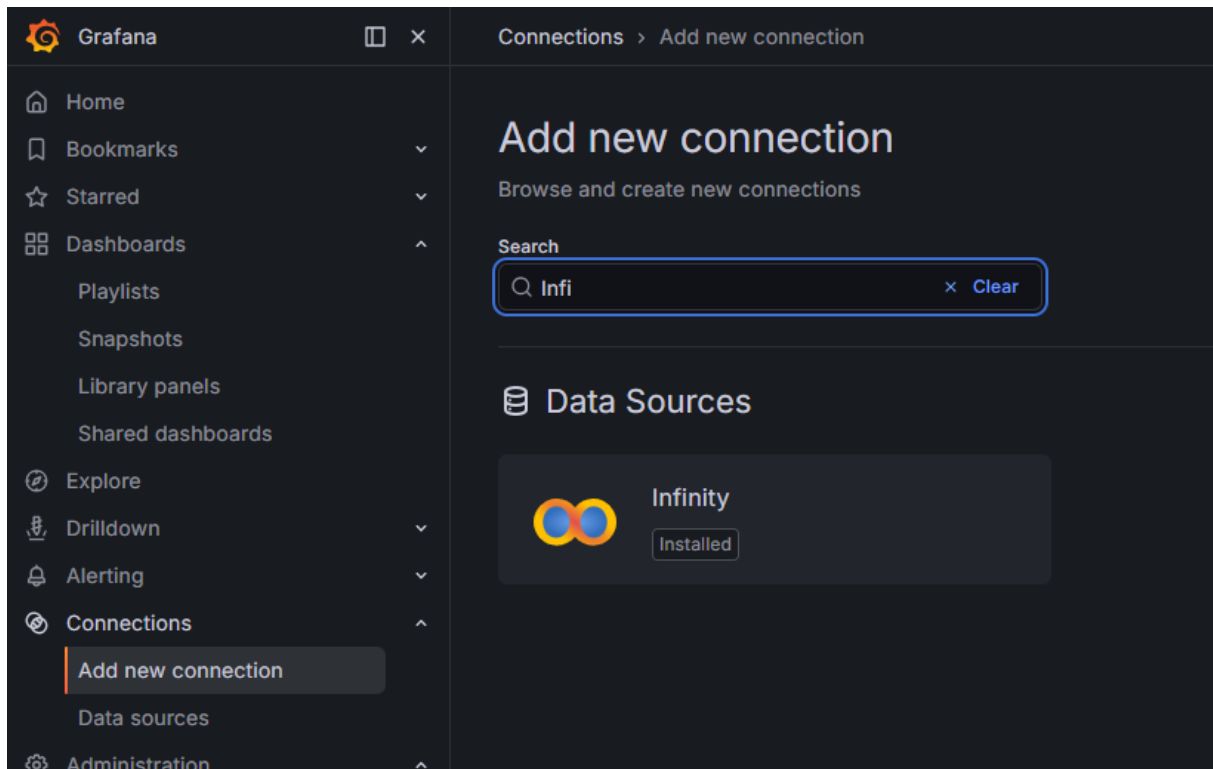
Purpose of this guide

The previous version of this guide described an integration that is deprecated, as the plug-in used for the integration is deprecated. This guide describes the integration mechanism via the Infinity data source.

Important information: If Grafana requests a large amount of data from the ESM server, this might cause performance and processing issues on the ESM server.

Requirements

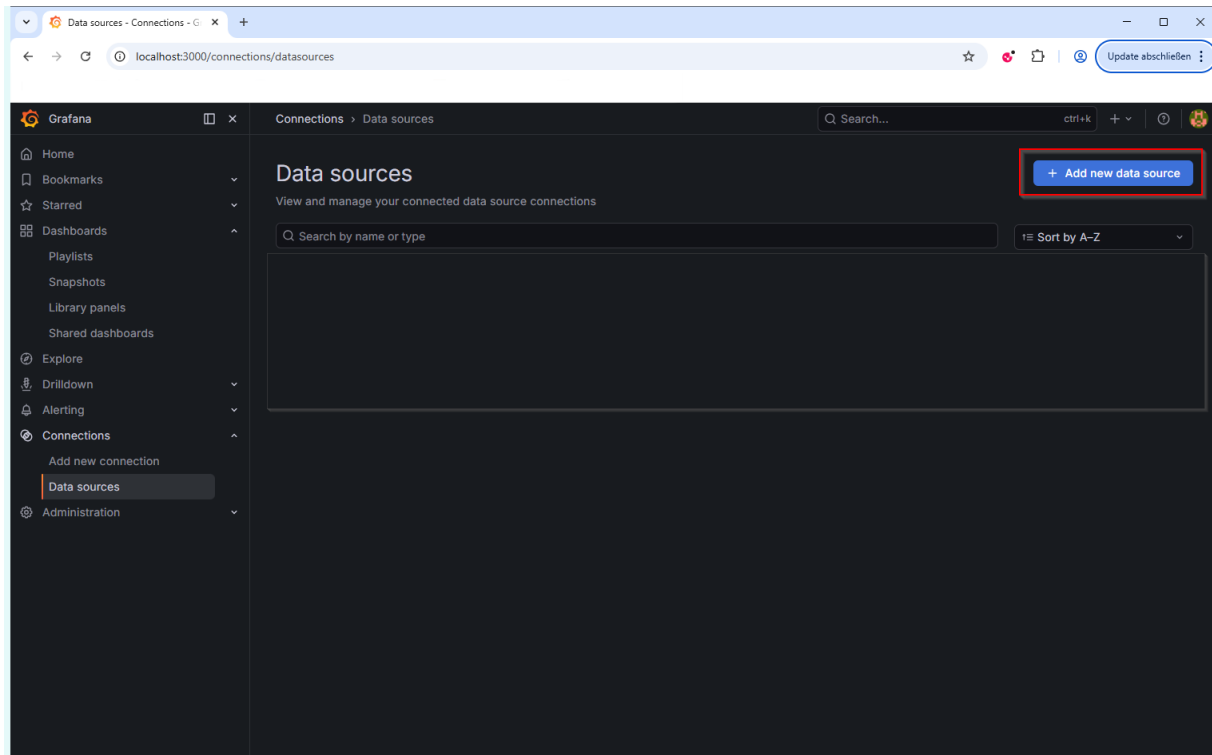
- ESM Version 5.7.0.0-000 or later (already installed)
- Grafana (already installed)
- The Grafana System must be able to access the ESM Server REST API via URL
- The Infinity plug-in for data source in Grafana must be installed.



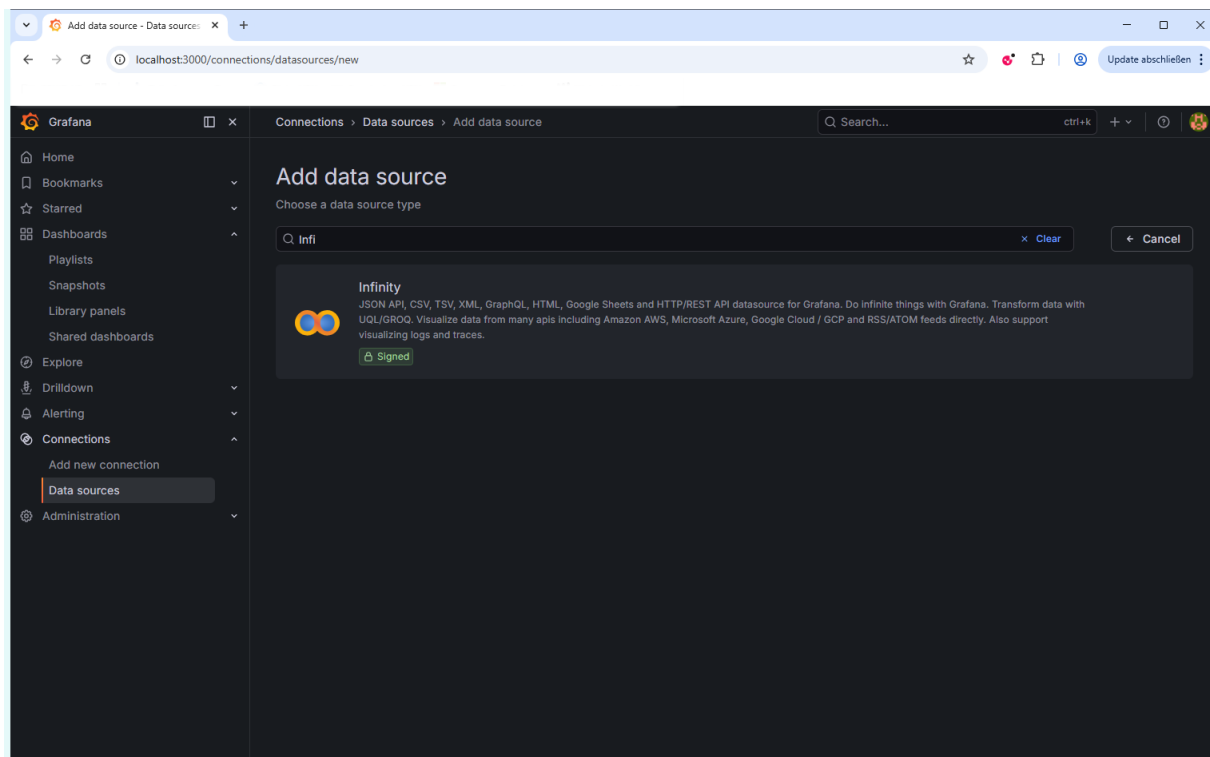
Step by step setup

1. Setup ESM data source in Grafana

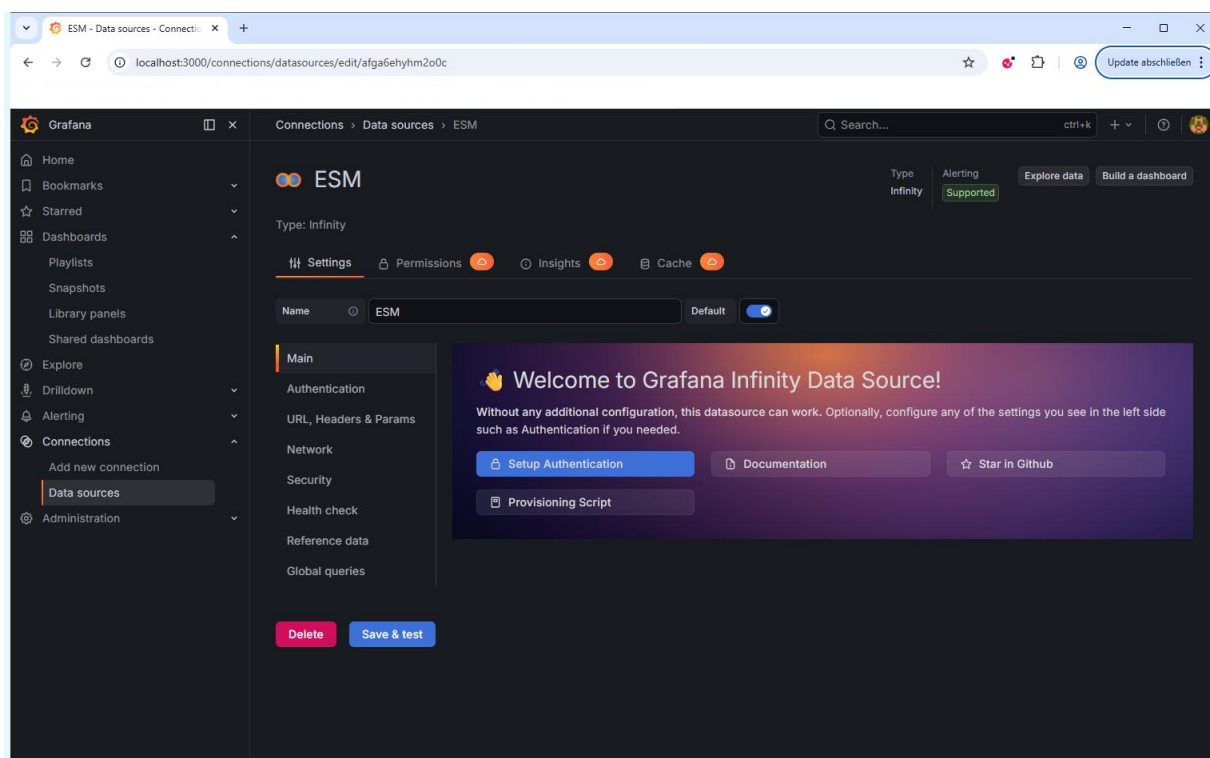
Open Connection -> Data sources and click Add new data source:



Start Typing Infinity in the search field to filter the available sources or search the Infinity source manually. Click on it to start the configuration of the data source:

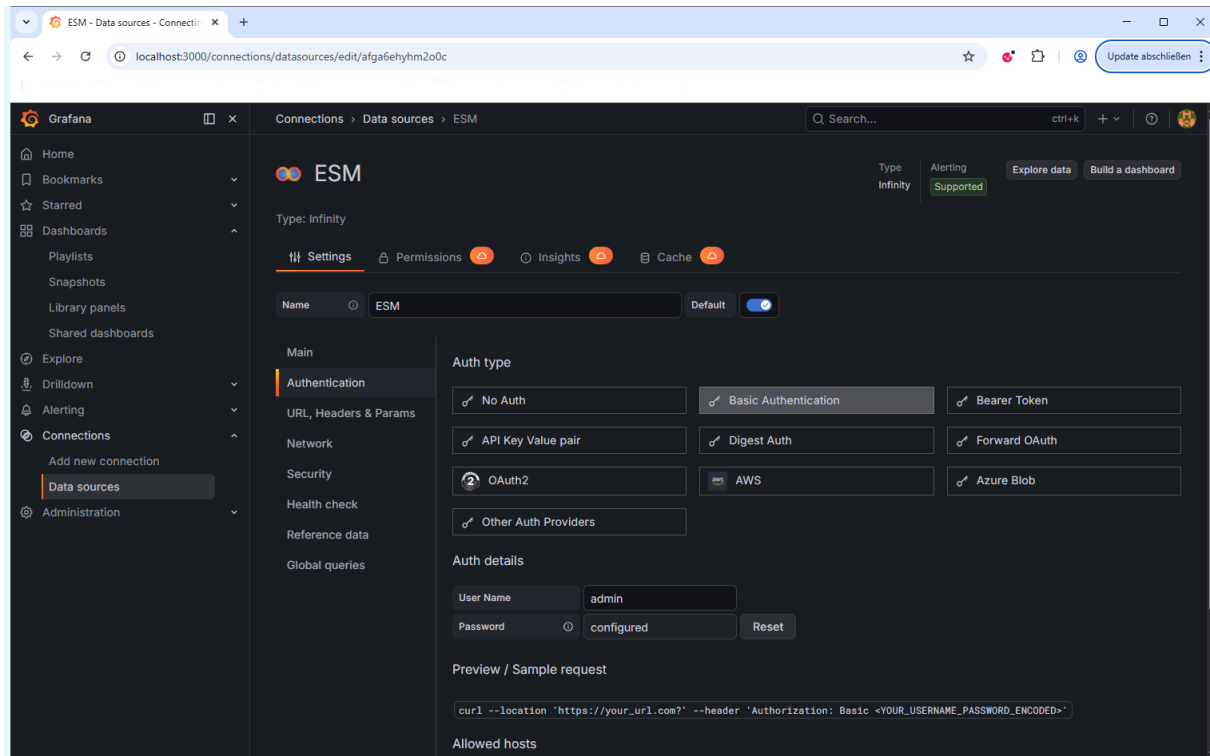


First, adjust the name to something that suits you, in this example ESM is used.

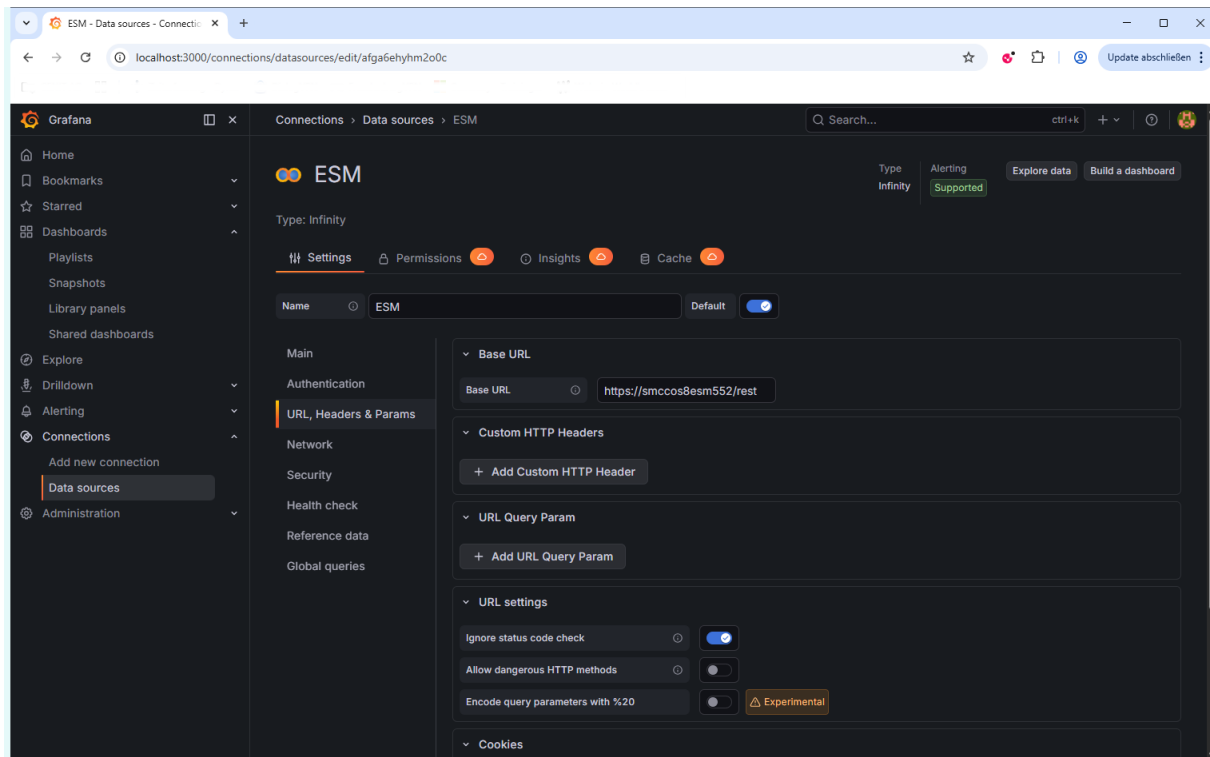


Switch to the Authentication part and specify Basic Authentication along with a user and password. In this example the admin account is used.

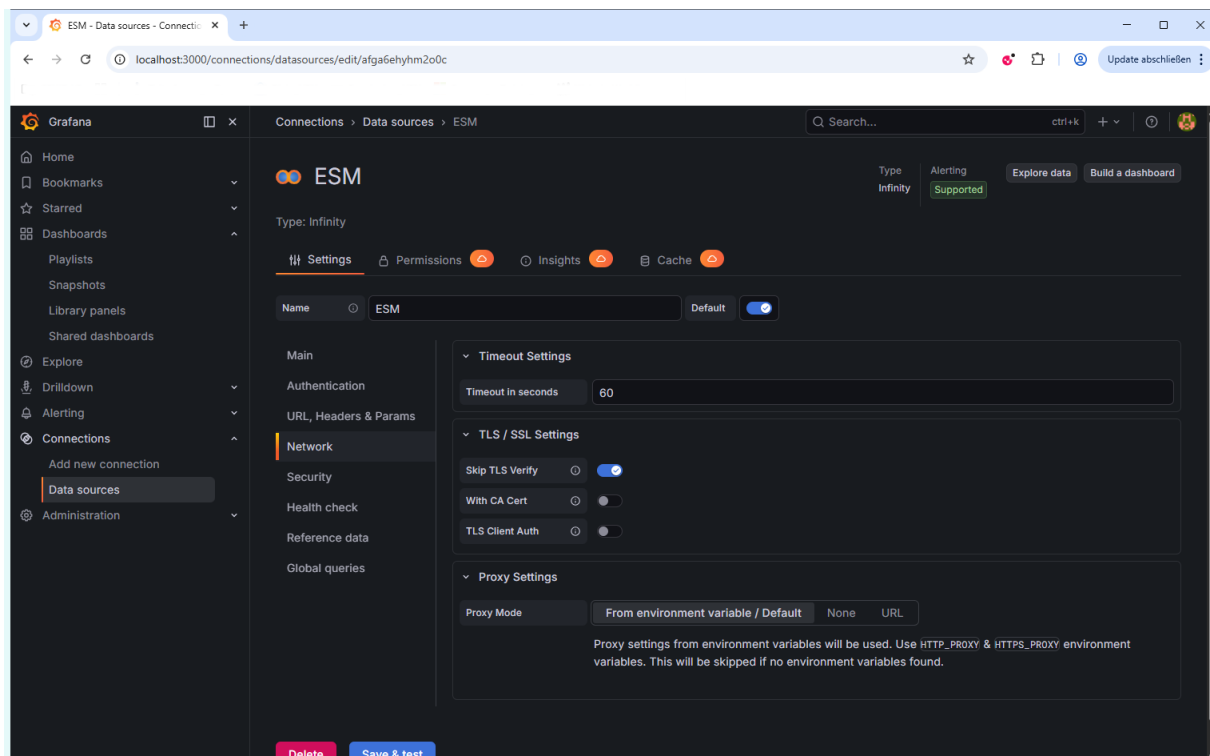
Note: Any user that has access to ESM is working, also users with role “guest”.



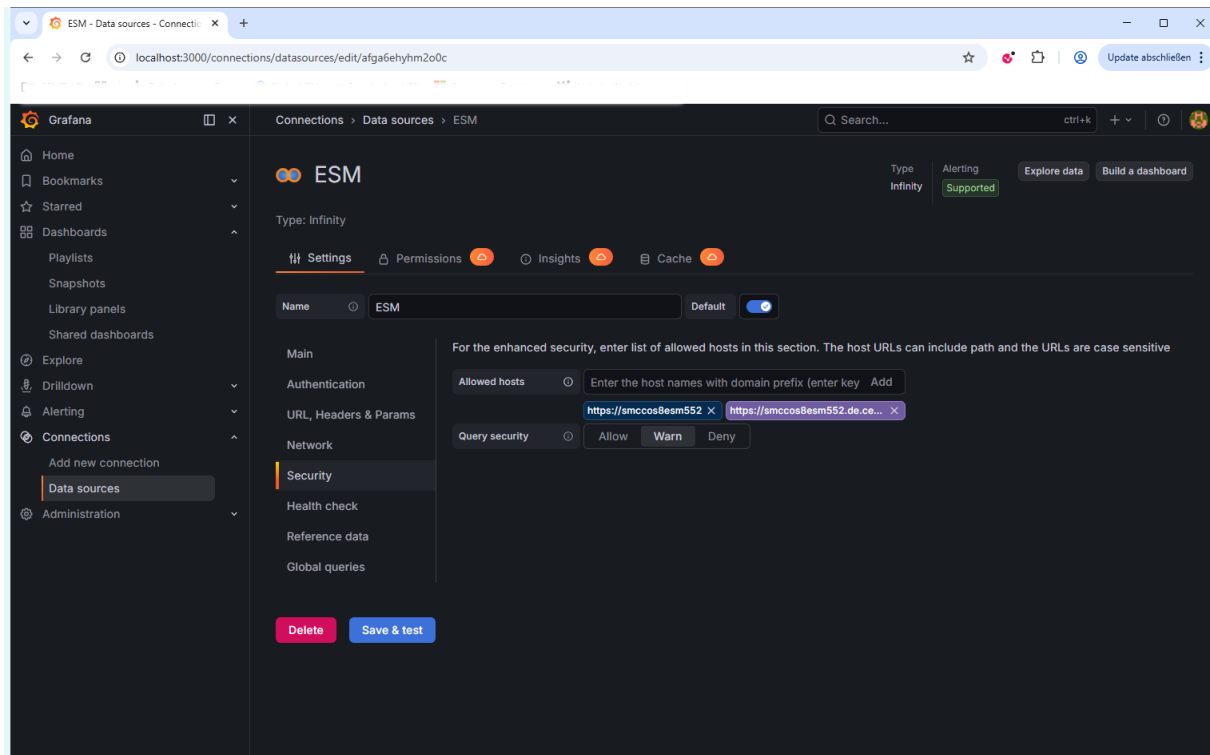
Browse to URL, Headers & Params. Enter the URL of your ESM server with /rest in Base URL and if not done by default, turn on “Ignore status code check”. In this example the URL is <https://smccos8esm552/rest>.



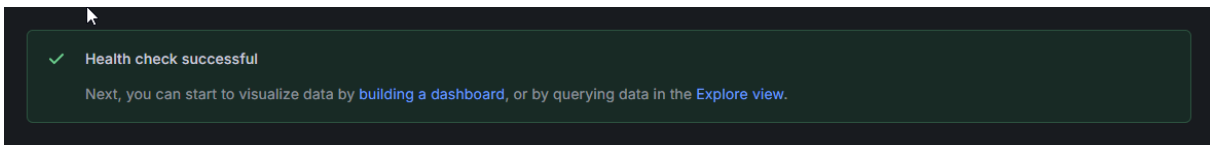
Optional – If self-signed certificate is used, go to Network and turn on Skip TLS verify



Optional - In Security, please add allowed hosts as URL. In this example the ESM server hostname and fully qualified name are used. Do the same in your system:

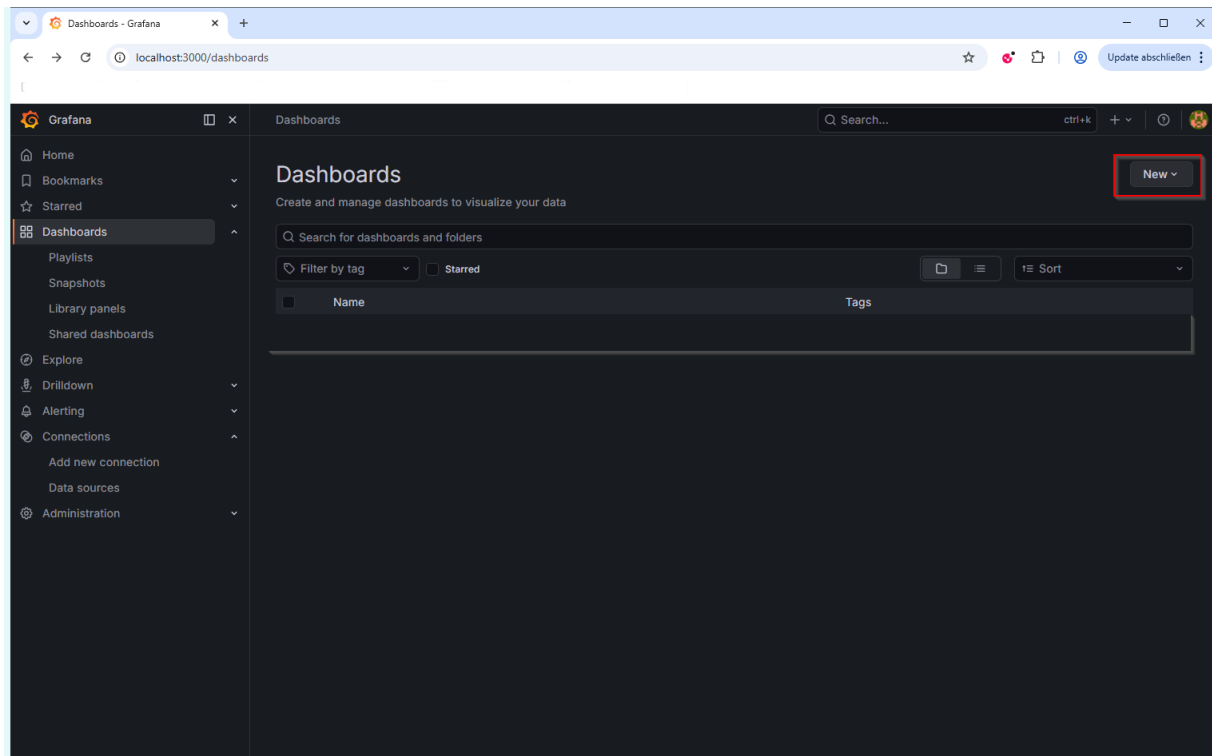


For Health Check, Reference Data and Global queries no adjustments are needed. Click on “Save & test” to save the source.

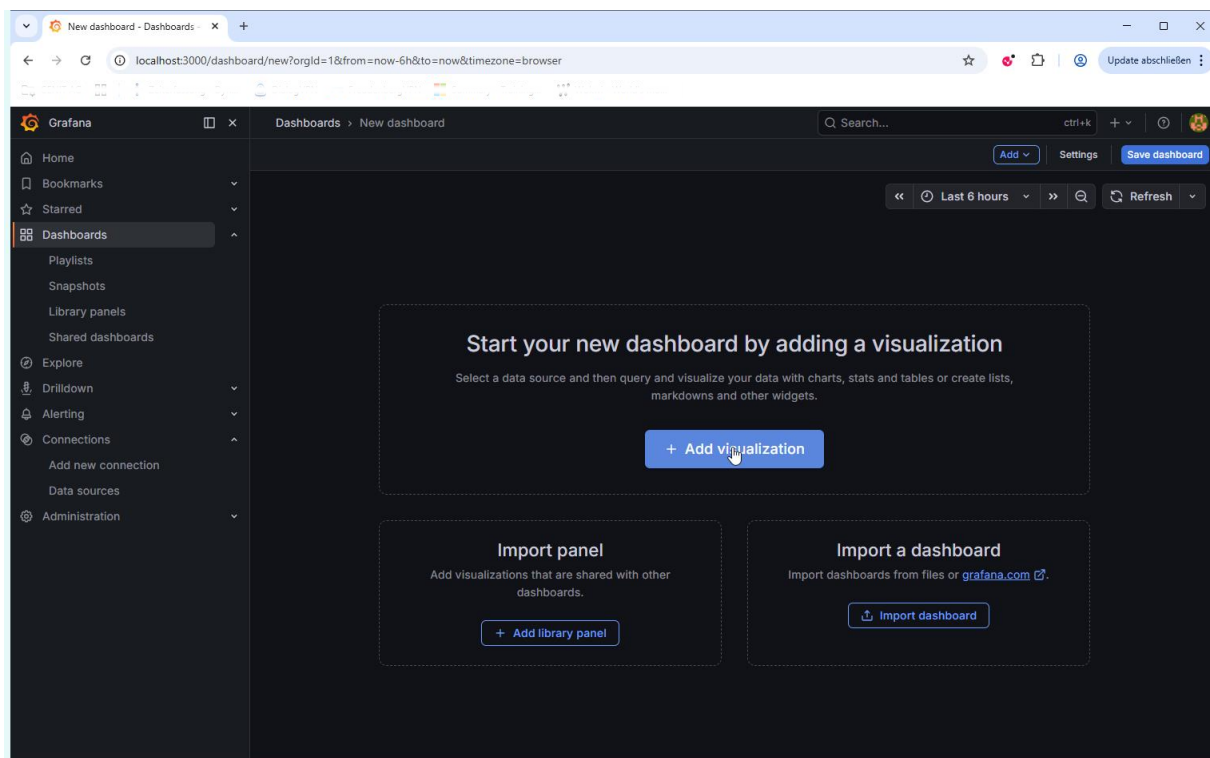


2. Setup of a dashboard – Preparation for Metrics

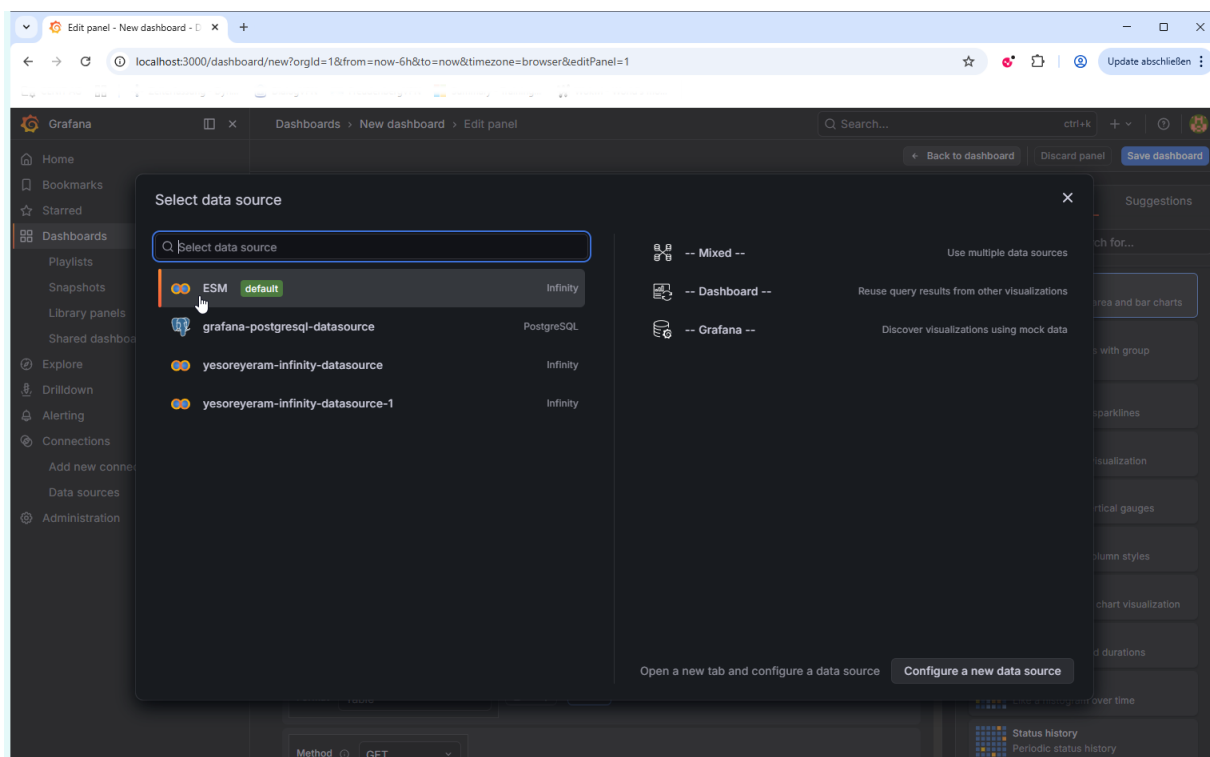
Select Dashboard from the side bar and click on New -> New Dashboard



Click on Add visualization:

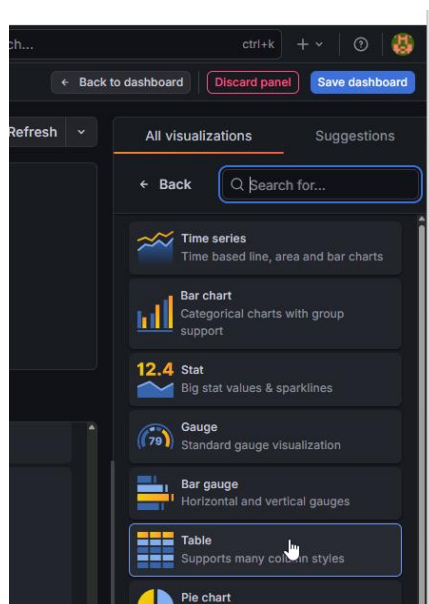


Click on the ESM data source that was created:

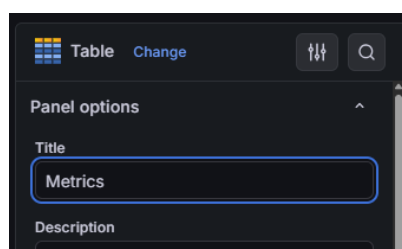


To show probe information in Grafana, the ID of the probe configuration is required in the setup for the individual views. For an easier setup process, the first view that we will add provides the ID of the probe configuration in ESM along with the names. This helps to find the ID for the relevant probes and makes it easier to set up the individual views.

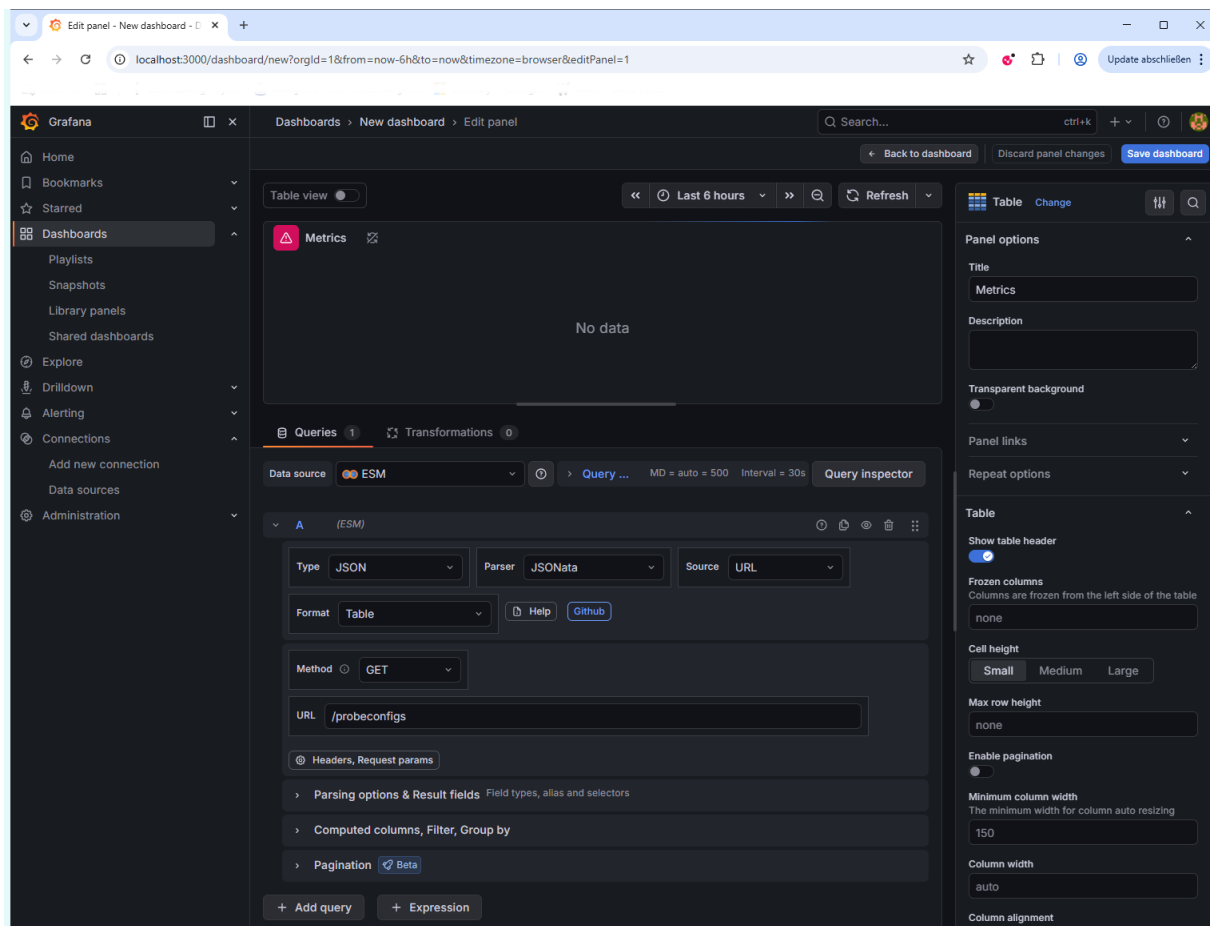
First select the table visualization from the right bar:



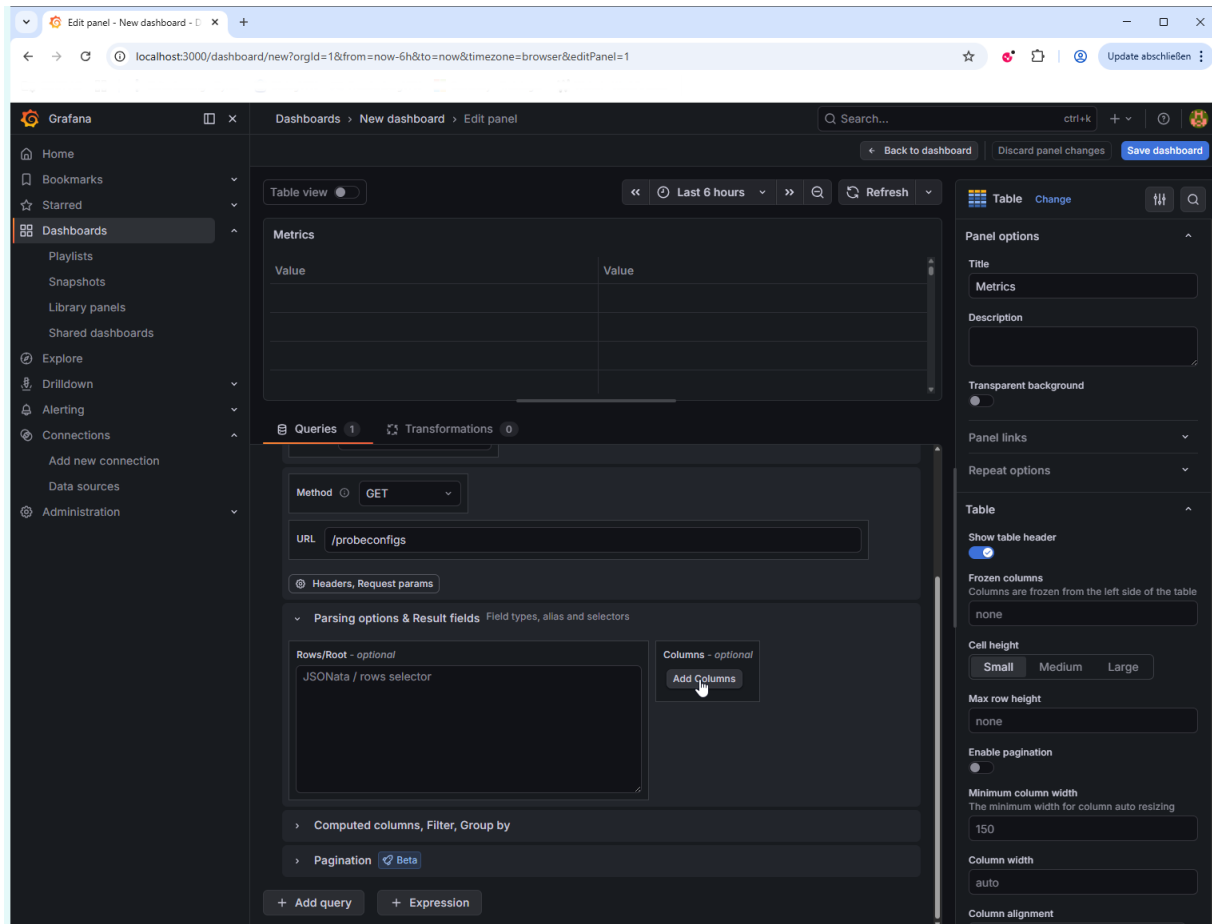
After changing it to table you can directly specify a title. Metrics will be used as an example for this panel:



In A, if not set up by default, please set Type JSON, Parser JSONata, Source URL and Format Table. As Method use GET and enter /probeconfigs in the URL:



Open the Parsing options & Result fields section and click on Add Columns:



As new column specify:



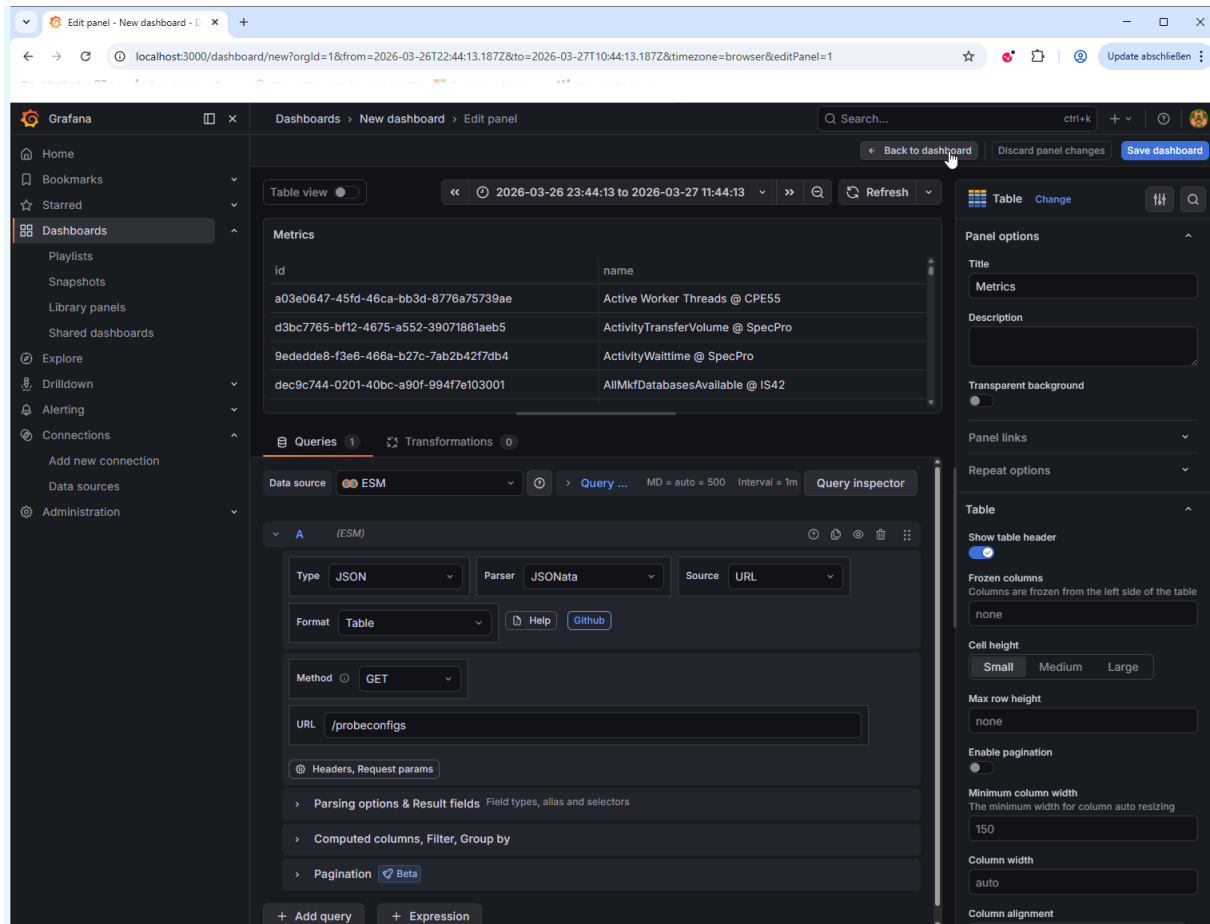
Click on Refresh in the panel – you should now see a table filled with information:

The screenshot shows the Grafana dashboard editor interface. The main panel displays a table view of metrics. The table has two columns: 'id' and 'name'. The data rows are as follows:

id	name
a03e0647-45fd-46ca-bb3d-8776a75739ae	Active Worker Threads @ CPE55
d3bc7765-bf12-4675-a552-39071861aeb5	ActivityTransferVolume @ SpecPro
9ededde8-f3e6-466a-b27c-7ab2b42f7db4	ActivityWaittime @ SpecPro
dec9c744-0201-40bc-a90f-994f7e103001	AllMkfDatabasesAvailable @ IS42

The interface includes a sidebar on the left with navigation options like Home, Bookmarks, Starred, Dashboards, Playlists, Snapshots, Library panels, Shared dashboards, Explore, Drilldown, Alerting, Connections, Add new connection, Data sources, and Administration. The top right shows the 'Edit panel' tab and a search bar. The bottom of the main panel has buttons for '+ Add query' and '+ Expression'. The right sidebar contains panel options such as Title, Description, Transparent background, Panel links, Repeat options, Table (with Show table header, Frozen columns, Cell height, Max row height, Enable pagination, Minimum column width, Column width, and Column alignment).

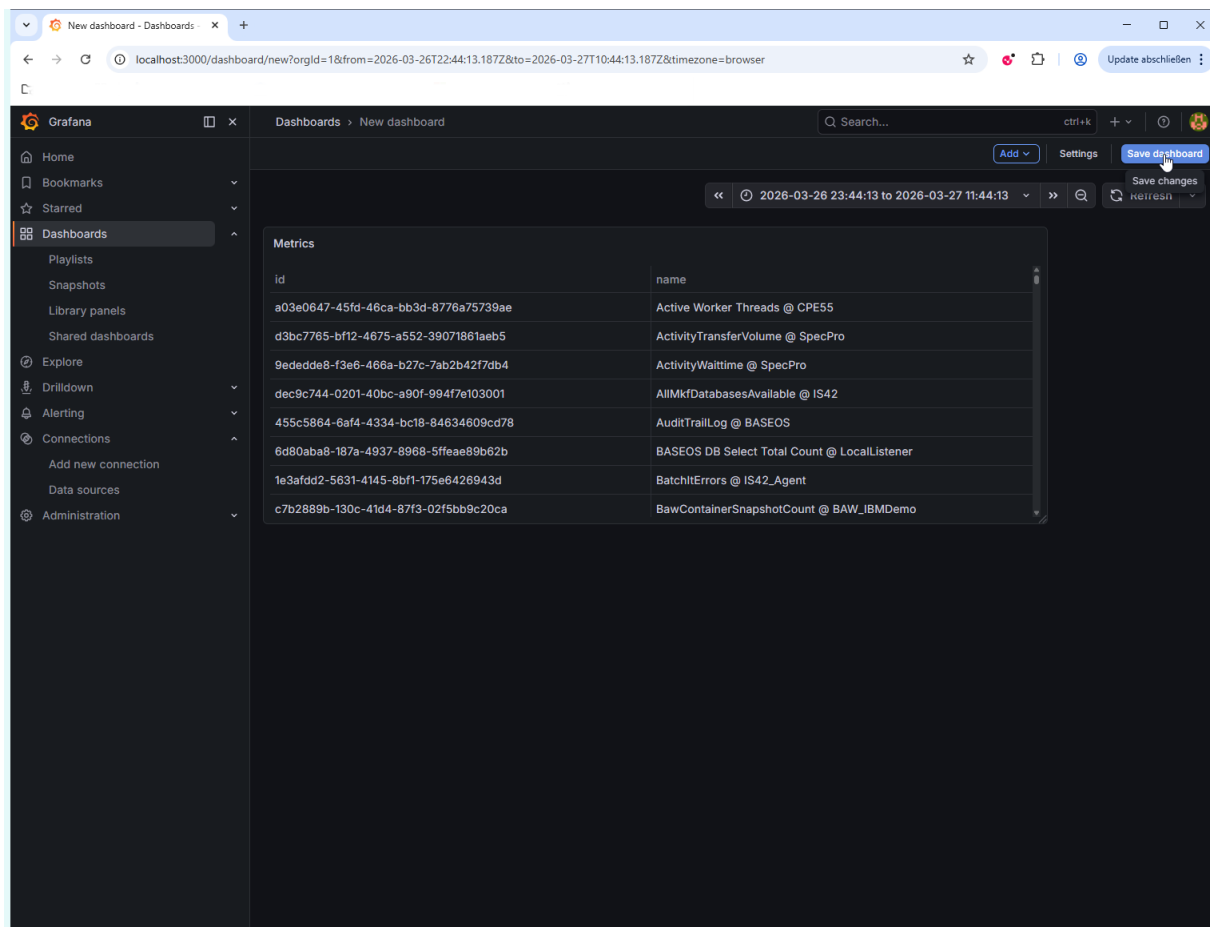
Click on Back to dashboard to switch into the dashboard:



You should now see this view in the dashboard:

Metrics	
id	name
a03e0647-45fd-46ca-bb3d-8776a75739ae	Active Worker Threads @ CPE55
d3bc7765-bf12-4675-a552-39071861aeb5	ActivityTransferVolume @ SpecPro
9ededde8-f3e6-466a-b27c-7ab2b42f7db4	ActivityWaittime @ SpecPro
dec9c744-0201-40bc-a90f-994f7e103001	AllMkfDatabasesAvailable @ IS42
455c5864-6af4-4334-bc18-84634609cd78	AuditTrailLog @ BASEOS
6d80aba8-187a-4937-8968-5ffeae89b62b	BASEOS DB Select Total Count @ LocalListener
1e3afdd2-5631-4145-8bf1-175e6426943d	BatchItErrors @ IS42_Agent
c7b2889b-130c-41d4-87f3-02f5bb9c20ca	BawContainerSnapshotCount @ BAW_IBMDemo

For adding additional views with sample information like value, message etc. we require this information. Therefore it is best to save the dashboard and duplicate the browser Tab. As title for the dashboard ESM – Metrics is used in this example.



Note: Instead of creating the metric panel you can also read the ID from the ESM Console URL when selecting a Probe in the sidebar.

← → ↻ https://smccos8esm552/ui/#/probeConfigs/122fec81-62e0-32c8-ad41-5be333c07ac2?f_name=CPU

CENIT AG Google Fragen zu dieser Seite stellen - smccos8esm552

⚙️

ESM 5.7.0.0-002

⋮

📊

🔧

🔊 3

👤

🔇

Probes

Cpu @ CPE5510_Agent

+ CPU ↻

Cpu @ Server_Agent
CPU

Cpu @ CMODSSL_Agent
CPU

Cpu @ ConMan_Agent
CPU

Cpu @ CPE5510_Agent
CPU

Cpu @ CPE557_Agent
CPU

Cpu @ DCAPAgent
CPU

✎ Edit

📄 Copy

🗑 Delete

▶ Execute

🔵 ✓ Active

CPU

Metric mode

Request Count: 5

Schedule & Automation

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2. Setup of a dashboard – Add ESM Metrics

You can now decide if you want to add the other visualizations to this dashboard or to an additional dashboard. In this example the Metrics dashboard is kept separate and a new dashboard for the Events will be added. Open a second browser tab and repeat the steps to add a new dashboard.

For creating a visualization panel, it is important to know what type of data you want to show in the panel.

For the data you can either use sample information or incident information.

Sample information are suitable for showing a time series graph or something equal.

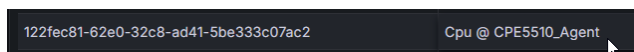
If you want to create an incident list (e.g. for a logfile) or only show the current value in a Gauge panel you can use the incident information.

This sample or incident data is based on the URL. Examples for both will be given in the next steps.

2.1. Setup of a dashboard – Add Sample Metrics (Time Series Graph)

As a first step the following describes the creation for a visualization of sample data like CPU information as a Time Series graph.

Once the visualization configuration is open, select Time Series from the right sidebar. As title you can specify the Probe Name for which you configure this visualization. (See Metrics screenshot below).

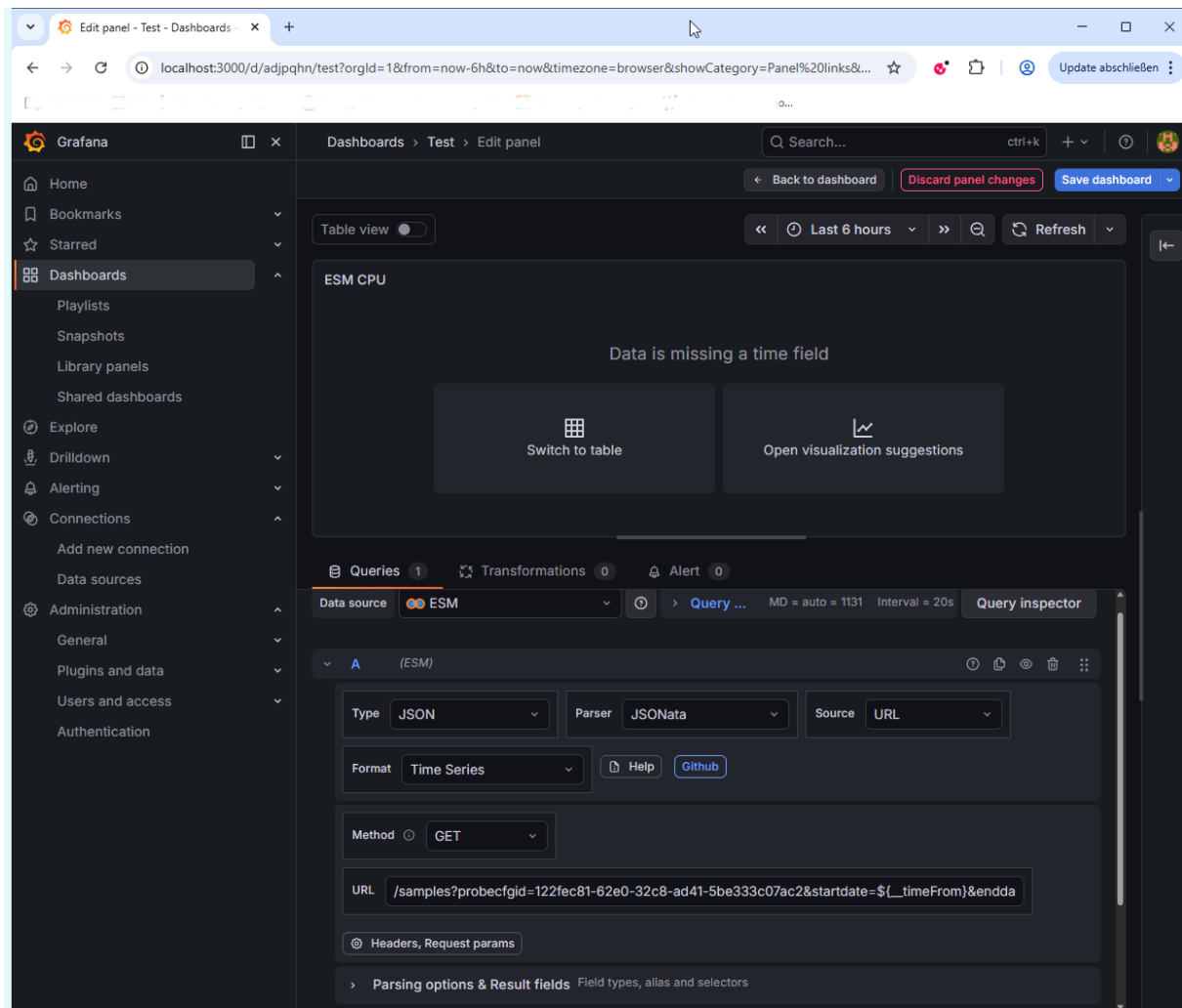


For Type keep JSON, for Parser JSONata and for Source URL. Change Format to Time Series. Keep GET as method and add a URL as follows:

```
/samples?probecfgid=122fec81-62e0-32c8-ad41-5be333c07ac2&startdate=${__timeFrom}&enddate=${__timeTo}
```

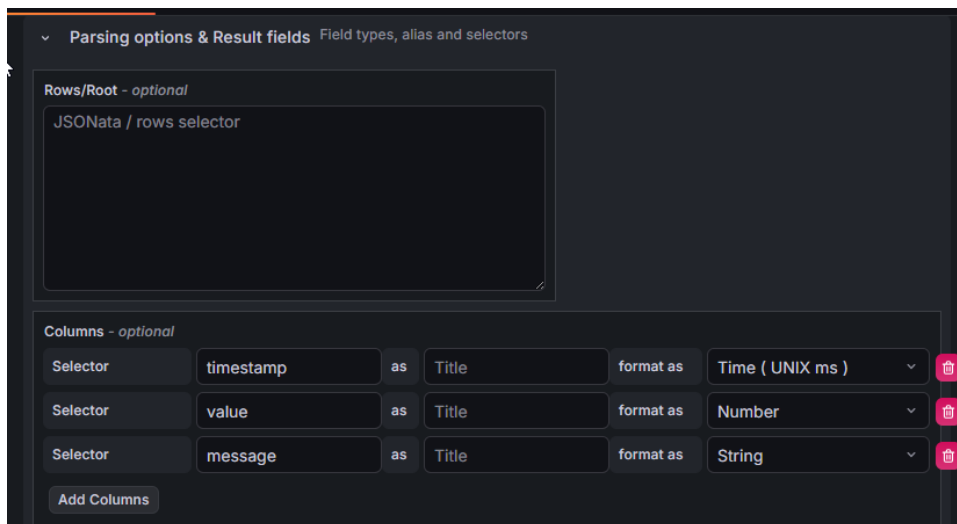
The `probecfgid` can be copied from the Metrics as well.

`&startdate=${__timeFrom}&enddate=${__timeTo}` is used to select only the data valid for the selection in the period of the panel.

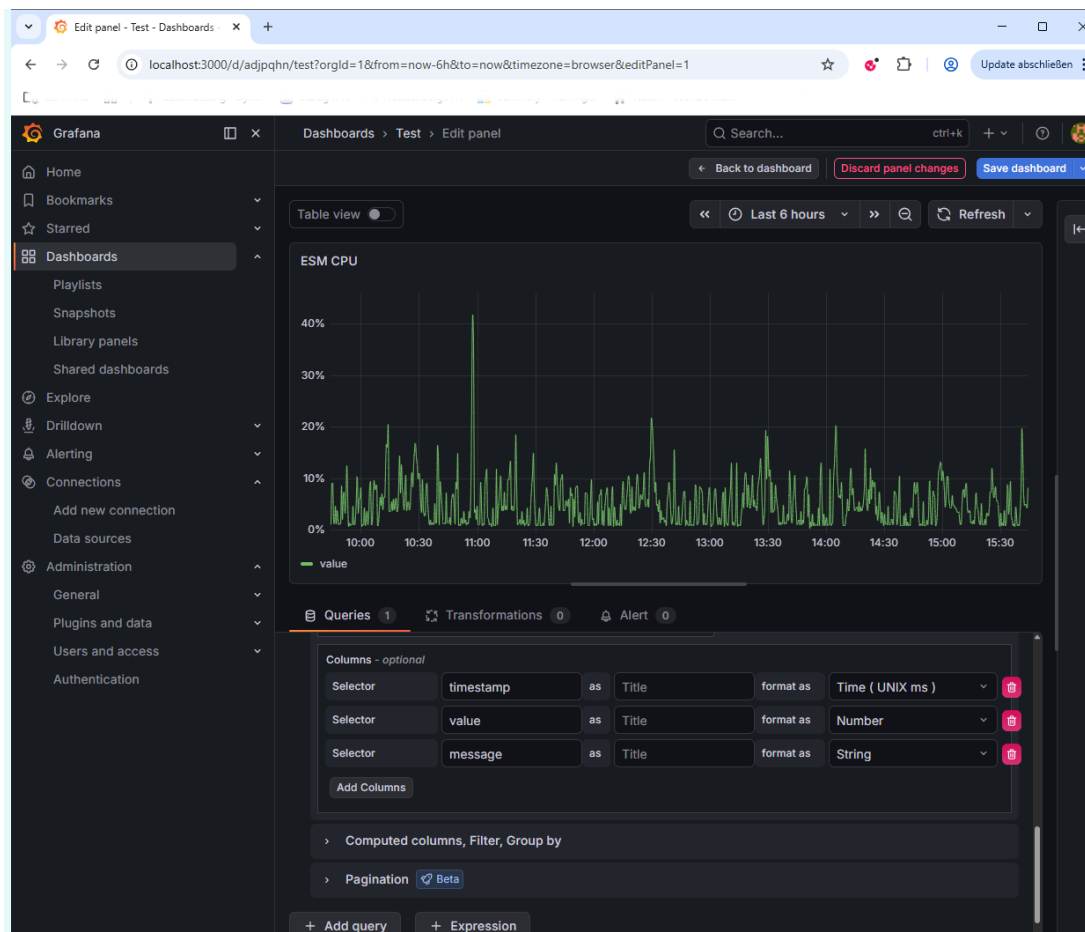


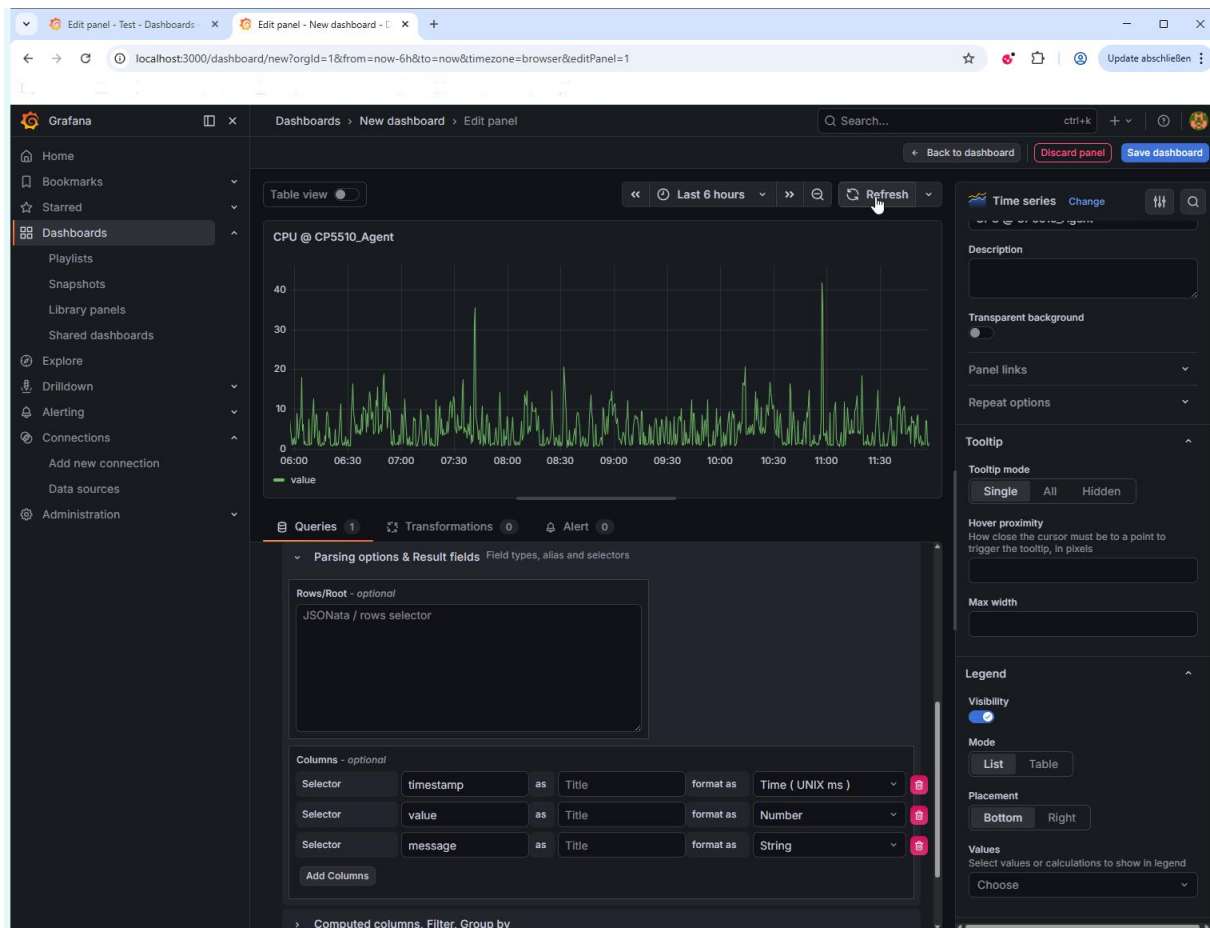
It is necessary to map sample information to Grafana, therefore open Parsing options & Result Fields and add the following columns.

Note: These are required to get useful data, others are available



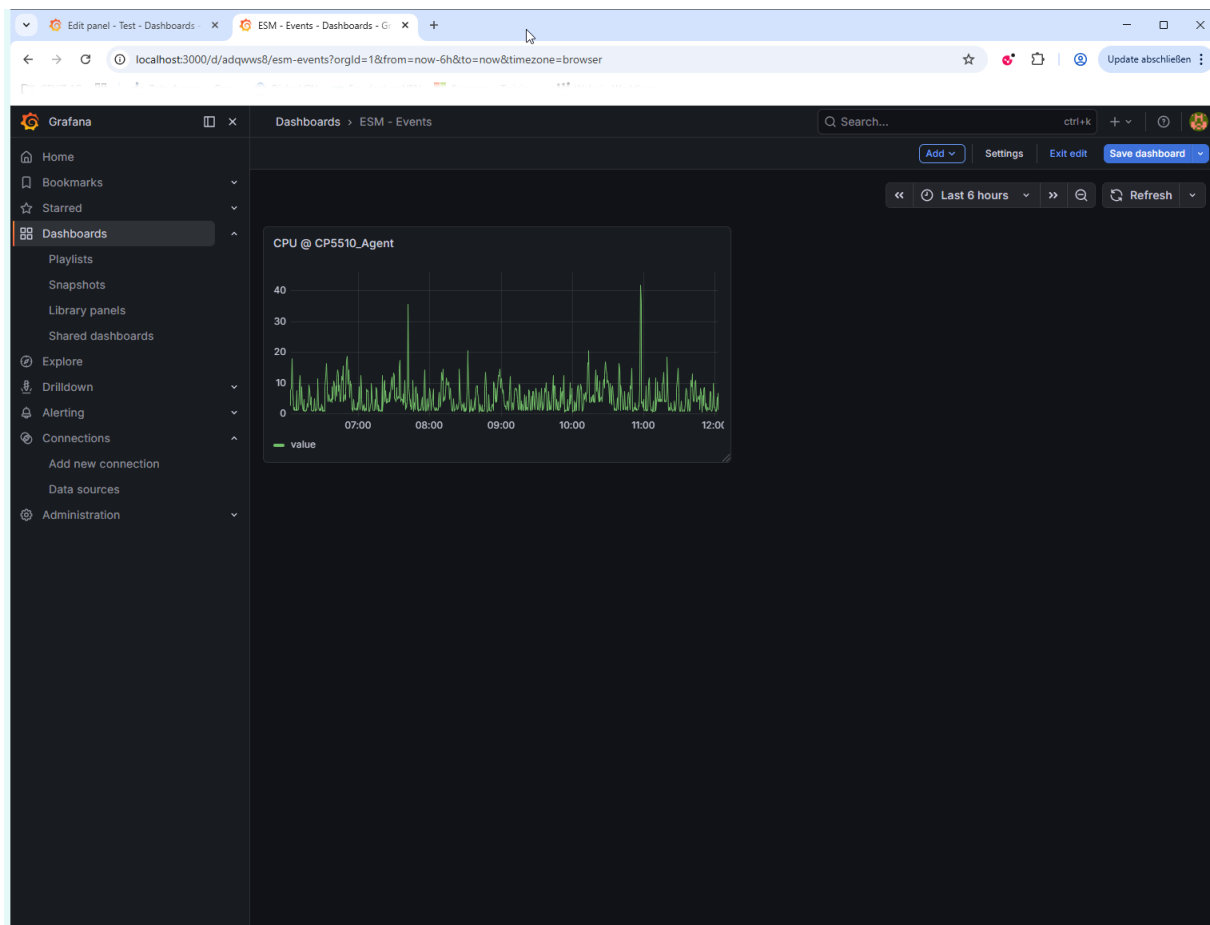
Click on Refresh in the Panel – A graph should now be visible:





You can click on Save dashboard and go back to the dashboard.

The panel is added. You can add additional panels with same layout using this method.



2.2. Setup of a dashboard – Add Incident Metrics (List of incidents for table visualization or single incident value e.g. for gauge visualization or status)

The second possibility is to add visualization based on incident data. This is typically useful for adding a single data information like e.g. current CPU info in a gauge or simple status for a resource. Also listing incidents from a logfile in a table can be covered. When using incident information, links to the ESM console can be created and added to the visualization.

The settings are almost the same as for the sample data, the URL differs slightly:

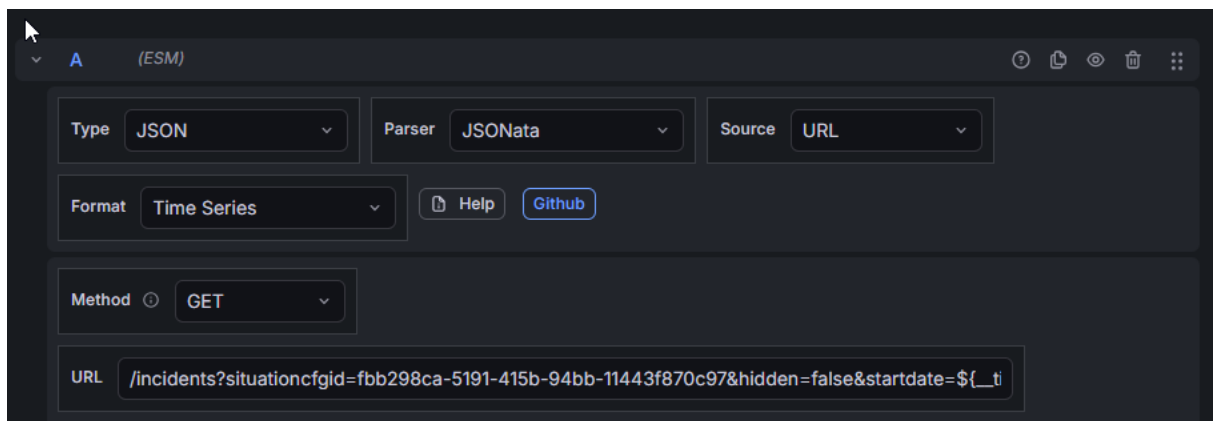
```
/incidents?situationcfgid=fbb298ca-5191-415b-94bb-11443f870c97&hidden=false&startdate=${__timeFrom}&enddate=${__timeTo}
```

The situationcfgid can be copied from the Metrics as well.

`&hidden=false` defines that closed incidents are not collected

`&startdate=${__timeFrom}&enddate=${__timeTo}` is used to select only the data valid for the selection in the period of the panel.

In this example, a table visualization that covers logfile incidents is created. For A the following is given:



In the Parsing options & Result fields the following columns are added:

Parsing options & Result fields Field types, alias and selectors

Rows/Root - optional

JSONata / rows selector

Columns - optional

Selector	Value	as	Alias	format as	Format	Action
timestamp	as	TimeStamp	format as	Time (UNIX ms)	🗑️	
value	as	Value	format as	Number	🗑️	
message	as	Message	format as	String	🗑️	
id	as	ID	format as	String	🗑️	

Add Columns

Note: When using the ID in the Parsing Options & Result fields, you can link the information directly to the incident in the ESM console. E.g. in this case the added link in the right bar below Data links looks like this:

[https://smccos8esm552/ui/#/subsystems?incident=\\${__data.fields.ID}](https://smccos8esm552/ui/#/subsystems?incident=${__data.fields.ID})

Edit link ✕

Title

ESM Dashboard Link

URL

https://smccos8esm552/ui/#/subsystems?incident=\${__data.fields.ID}

Open in new tab ☒

Cancel Save

Furthermore, there are possibilities in the right bar to do colorings etc.

In this example the table that is created looks like this:

Logfile @ CPE5510WASLOG

TimeStamp ↓	Value	Message	ID
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540587
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540586
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540585
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:8815:43fb:946b%6:389"	540584
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540583
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540582
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540581
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540580
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540579
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540578
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540577
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540576
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540575
2026-03-30 07:21:08	30	UserRegistryC SECJ0281E - Error creating user registry object. The exception is java.lang.NumberFormatException: For input string: "0:0:0:b09b:88	540574

Here is one more example for gauge visualization:

This is the URL for a CPU incident:

```
/incidents?situationcfid=122fec81-62e0-32c8-ad41-5be333c07ac2&hidden=false&startdate=${__timeFrom}&enddate=${__timeTo}
```

The URL for the link to the incident in ESM looks like this:

[https://smccos8esm552/ui/#/subsystems?incident=\\${__field.labels.ID}](https://smccos8esm552/ui/#/subsystems?incident=${__field.labels.ID})

Note: The \${} Variable for ID is different – this is something that comes from Grafana. This is not influenced by ESM.

Edit link

Title

ESM

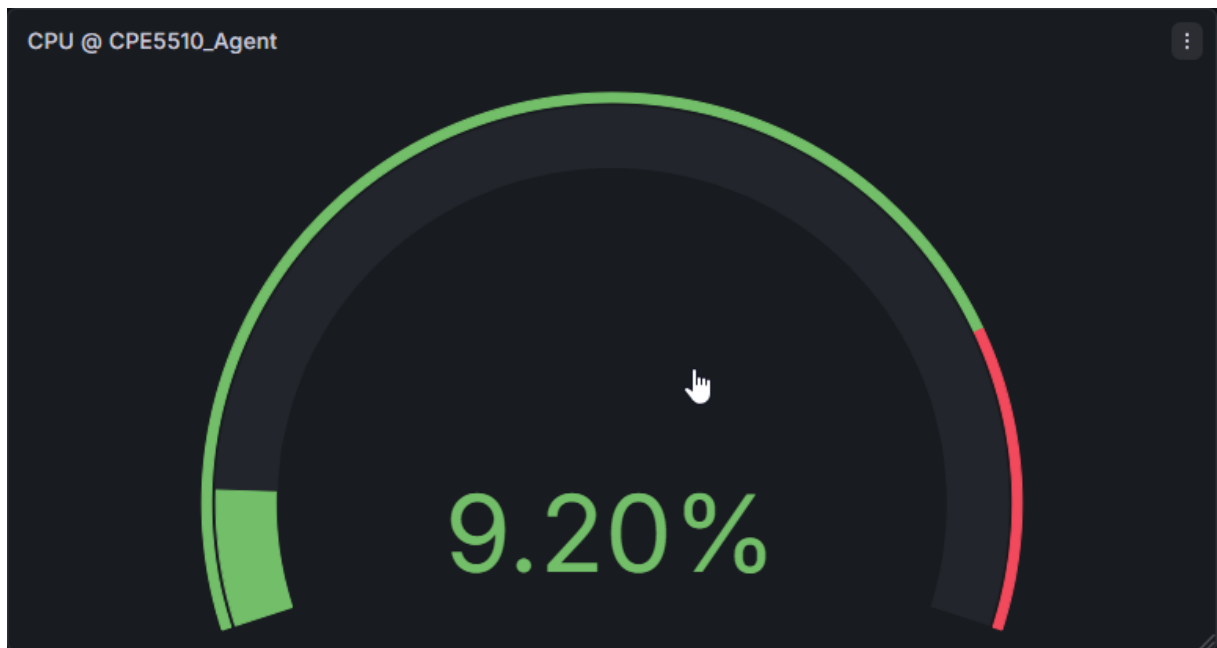
URL

https://smccos8esm552/ui/#/subsystems?incident=\${__field.labels.ID}

Open in new tab

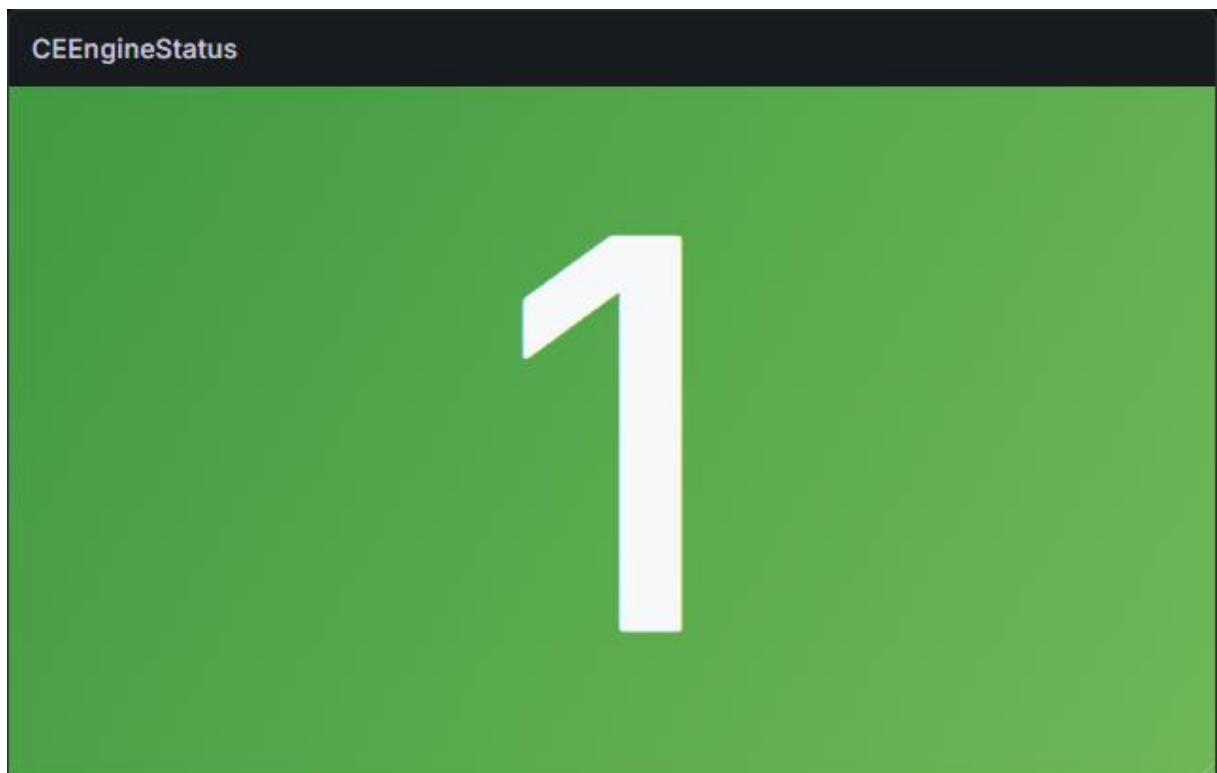
☒

Cancel Save



Example for a Stat visualization using this URL:

```
/incidents?situationcfgid=96fefe34-4dfe-4bba-b3c7-7f90ea7902f2&hidden=false&startdate=${__timeFrom}&enddate=${__timeTo}
```



Contact Information

If you have any questions, please contact us at ECM.SystemMonitor@cenit.com.

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